Md. Reazul Islam

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Objective

Ph.D. student in Computer Science at the University at Albany, SUNY, specializing in Artificial Intelligence, Deep Learning, Computer Vision, and Smart Healthcare. My doctoral research focuses on motion-aware, domain-adapted deep learning and AI for non-contact multi-vital and neurological monitoring of neonates in the NICU. My long-term goal is to develop robust, clinically relevant AI systems for continuous monitoring of heart rate, respiratory rate, motion, and neurological status. I aim to advance trustworthy, scalable, and non-invasive healthcare technologies that improve patient safety, support early detection, and strengthen intelligent healthcare systems in the United States and globally.

Education

- **M.Sc. in Computer Science and Engineering** Jan 2015 – Dec 2015
  Jahangirnagar University (JU), Bangladesh
- **B.Sc. in Computer Science and Engineering** Jan 2011 – Dec 2014
  Jahangirnagar University (JU), Bangladesh

Professional Experience

- **Graduate Teaching Assistant** Aug 2025 – Continue
  University at Albany, SUNY
- **Assistant Professor, CS** Jan 2024 – Aug 2025
  American International University-Bangladesh
- **Lead Researcher (IoT, AI & ML)** Jan 2023 – Jan 2026
  Advanced Machine Intelligence Research Lab
- **Lecturer, CS** Feb 2022 – Jan 2024
  Bangladesh University of Business and Technology
- **Lecturer, CS** Jan 2017 – Apr 2018
  Daffodil International University
- **Team Lead, Android & Web** May 2018 – Mar 2021
  GSPI LTD, Dhaka
- **Web Developer** Jan 2014 – Feb 2017
  Techideasolutions

Publications

Journals:

1. **Md.Reazul Islam**, Khondokar Oliullah, Md Mohsin Kabir, Dr. M. F. Mridha, Ashifur Rahman, Muhammad Fayyaz Khan. "Machine Learning-Driven IoT Device for Women Safety: A Real-Time Sexual Harassment Prevention System". *Journal of Multimedia Tools and Applications.*
2. **Md.Reazul Islam**, Khondokar Oliullah, Md Mohsin Kabir, Munzirul Alom, Dr. M. F. Mridha. "Machine Learning Enabled IoT System for Soil Nutrients Monitoring and Crop Recommendation". *Journal of Agriculture and Food Research.*
3. **Md.Reazul Islam**, Md. Mohsin Kabir, M. F. Mridha, Sultan Alfarhood, Mejdl Safran, Dunren Che. "Deep Learning-based IoT System for Remote Monitoring and Early Detection of Health Issues in Real-Time". *Journal of Sensors.*
4. Khondokar Oliullah, **Md.Reazul Islam**, Jahirul Islam Babar, et al. "FruVeg_MultiNet:A Hybrid Deep Learning-Enabled IoT System for Fresh Fruit and Vegetable Identification with Web Interface and Customized Glass for Visually Impaired Individuals". *Journal of Agriculture and Food Research.*
5. Md.Owafeeuzzaman Patwary, Apurbo Biswas, **Md.Reazul Islam**, Md. Taufiqur Rahman,Syeda Asrafa Islam,Md. Abdullah-Al Jubair,Md Rakibul Islam, M. Firoz Mridha Ph.D. et al. "An Interpretable Deep Learning Approach for Breast Cancer Classification and Tumor Localization from Histopathological Images". *Journal of Results in Engineering.*

6. Khondokar Oliullah, Mahedi Hasan Rasel, Md Manzurul Islam, **Md.Reazul Islam**, Md Anwar Hussen Wadud, Md Whaiduzzaman, et al. "A Stacked Ensemble Machine Learning Approach for the Prediction of Diabetes". *Journal of Diabetes & Metabolic Disorders*.

7. Md Owafeeuzzaman Patwary, **Md.Reazul Islam**, Nur-e Sarjina Khan Abtahi Islam, Md Abdullah-Al–Jubair, Md Jakir Hossen, MF Mridha, et al. "The ethics of artificial intelligence technology in academic work: assessing the line between assistance and plagiarism". *Journal of IJECE*.

8. **Md.Reazul Islam**, Khondokar Oliullah, Rajarshi Roy Chowdhury, Shaikat Das Joy, MM Fazle Rabbi, et al. "IntelligentGuard: Smart Doorbell with Deep Learning for Secure User Recognition and Instant Notifications". *Journal of IJEER*.

9. **Md.Reazul Islam**, Md Owafeeuzzaman Patwary, SM Ashiqul Islam, et al. "AI-Driven Internet of Things (IoT) dataset for remote health monitoring and fall detection in elderly people".*Elsevier*.

Conferences:

1. Ali Azgar, Md Imran Nazir, Afsana Akter, Md Saddam Hossain, Md Anwar Hussen Wadud, **Md.Reazul Islam**, et al. "MNIST handwritten digit recognition using a deep learning-based modified dual input convolutional neural network (DICNN) model". *ICICT-2024*.

2. TM Amir-Ul-Haaue Bhuivan, Md Anwar Hussen Wadud, Nazmunnahr Amie, Md Mahbub-Or-Rashid, TM Ahkam-Ul-Haque Bhuiyan, Jannatun Nyeem Jui, Anichur Rahman, **Md.Reazul Islam**, et al. "Brain tumor detection by image segmentation using customized unet deep learning based model". *ICEEICT-2024*.

3. **Md.Reazul Islam**, Arman Hossain, Sayefa Arafah, MM Fazle Rabbi, Khondokar Oliullah, et al. "A Portable Diagnostic and Medication System for Rural Areas Using IoT". *TEHI-2023*.

Book Chapters:

1. **Md.Reazul Islam**, Mirza Asif Asif Mahmud, Kazi Tanvir, Shaikat Das Joy, Mahfujur Rahman, Noboranjan Dey, Dipta Gomes, et al. "IoMT Architecture for Healthcare: Integration, Security, and Real-Time Decision Support". *IGI Global*.

2. Shaikat Das Joy, Noboranjan Dey, **Md.Reazul Islam**, Mirza Asif Mahmud, Mahfujur Rahman, Kazi Tanvir, Dipta Gomes, et al. "Synthetic Healthcare Data in Generation, Validation, and Integration for Ethical and Scalable Health Innovation". *IGI Global*.

3. **Md.Reazul Islam**, Khondokar Oliullah, MD Mahbub-Or-Rashid, Jugal Krishna Das, et al. "Design and Implementation of a Cost-Effective Automatic Water Pump Control System for Domestic Applications Using IoT". *Intelligent Networks and Systems*.

4. Mirza Asif Mahmud,Mahfujur Rahman,Dipta Gomes,Kazi Tanvir,**Md.Reazul Islam** et al. "A Comprehensive Study on the Ethical Perspective of Integration of Agentic AI for Autonomous Vehicles". *IGI Global*.

5. Kazi Tanvir,Dipta Gomes,Mahfujur Rahman ,Noboranjan Dey,**Md.Reazul Islam** et al. "Vision Language Models in Healthcare Through a Multimodal Approach to Medical Imaging and Clinical Applications". *IGI Global*.

Research Presentations and Recognition at University at Albany, SUNY

- Presenter, UAlbany Showcase 2026, "AI–IoT Driven Edge–Cloud Collaborative Framework for Real-Time Air Pollution Monitoring and Mapping for Sustainable Urban Planning"
- Presenter, UAlbany Showcase 2026, "Closed-Loop Metabolic Monitoring: An IoT-Enabled Adaptive Framework for Personalized Dietary Intervention"
- Presenter, AI+ Symposium, University at Albany, "The Ethics of Artificial Intelligence in Academic Work: Assessing the Line Between Assistance and Plagiarism."
- Featured Research by UAlbany

Technical Skills

Coding: C, C++, Python, Java, Flutter, .NET Zero, PHP, JavaScript.
IoT Tools: Arduino, Raspberry Pi, Ubidots, ThingSpeak, Blynk.
ML-DL Tools: Keras, TensorFlow, Pandas, NumPy, PyTorch, Matplotlib.
Database: MySQL, SQLite, JDBC, MongoDB.

Selected Funded Projects with web link

- Design and Implementation of Cost-Effective Automatic Water Pump Controlling System.
- An Efficient IoT Enabled Smart chair with provision of sitting posture monitoring.

- ML-Enabled IoT Device for Soil Nutrients Monitoring and Crop Recommendation.
- Deep Learning-based IoT System for Remote Monitoring and Early Detection of Health Issues in Real-Time.
- A Novel ML-Supported IoT Device for Women's Self-Security System to reduce sexual harassment.
- Smart Shoes with IoT ML for Health Monitoring and Customized Food Chart Based on BMI

Awards and Prizes

- National Innovation Award, Digital Bangladesh Mela 2023 By Bangladesh Govt.
- Huawei ICT Incubator Innovation Award, 2022.
- Bangabandhu Innovation Grant, International Innovation Award, 2022.
- National Innovation Award, ICT Ministry of Bangladesh, 2019.
- Best Supervisor Award, Daffodil International University ICT Carnival, 2018.
- Best Teacher Award, Bangladesh University of Business and Technology.

Certification Courses

- Android Flutter and Dart (Udemy)
- Java Spring Framework and PHP (Durga Soft India)
- Embedded System/IoT Based (Pantech.ai)
- Python (Code with Harry),
- Machine Learning (Kaggle)

Leadership Skills

- Judge, Tri-County Science and Technology Fair 2026, USA
- Advisor, District Level Innovation Competition, ICT Ministry Bangladesh.
- Team lead, developer Schedule App and Publicity Subcommittee for ICPC Asia Dhaka Regional Contest.
- Advisor, IEEE BUBT Student Branch.
- Advisor, IEEE AIUB Student Branch.

Journal Referee

- Springer Nature, Discover Artificial Intelligence: 2025 to continue
- Springer Nature, Scientific Reports: 2025 to continue
- Springer Nature, Discover Computing: 2025 to continue
- Springer Nature, Cluster Computing: 2025 to continue
- Elsevier, Industrial Crops and Products: 2025 to continue
- IEEE Transactions on Multimedia: 2026 to continue
- IEEE Transactions on Circuits and Systems for Video: 2026 to continue
- Engineering Applications of Artificial Intelligence 2026 to continue

Media Coverage

- Showcase 2026: UAlbany Student Develops App to Help You Avoid Pollution Hot Spots — University at Albany
- The quest for an 'affordable' dress that detects sexual harassment
- Machine learning-driven IoT device for women's safety: a real-time sexual harassment prevention system
- Smart Learning Management System
- EasySchool virtual platform

Research Profile

Google Scholar— LinkedIn —IoT Lab — Portfolio